Bedrock Domain Scenario

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Integration Architecture

Instructions

Presentation Time : **20 minutes**
Q & A : **25 minutes**
Immediate Feedback : **15 minutes**

For the hypothetical scenario described below, you will need to document and present your proposed solution design followed by a Q&A session with a judge.

As part of the solution you should:

- Describe an appropriate solution that meets the business requirements
- Deliver all of the required deliverables as part of that presentation
- Address trade-offs and considerations for your solution options

You should target the presentation to the customer's Enterprise Architecture Group.

In cases where requirements are not explicitly stated for the scenario, you should use your best judgment and make appropriate assumptions based on the information provided and that in the Bedrock Overview (Domain Zero). You should indicate any assumptions that were made when designing the proposed solution. You will not have an opportunity to ask clarifying questions related to the hypothetical scenario. You will be evaluated on your ability to assess the scenario requirements, design a solution, communicate the proposed architecture, and justify the design decisions.

The executive will evaluate your solution in the following areas:

Objective	**Weighting**
Able to identify business requirements and understand the needs and challenges of a scenario	5%
Able to leverage the features and capabilities of the platform appropriately	10%

within the solution	
Able to clearly articulate the solution design	5%
Able to select the correct integration patterns for each use case and justify the solution	20%
Able to describe the key security considerations for designing and developing cloud based integrations	5%
Able to describe the various Salesforce APIs (e.g., REST, SOAP, Bulk, Streaming) and explain the most appropriate scenarios for each	15%
Able to describe event-driven integration patterns using Platform Events and Change Data Capture	10%
Able to describe how real-time and asynchronous callouts can be implemented declaratively (e.g., Flow HTTP Callout) and programmatically (Apex)	10%
Able to describe data virtualization patterns with Salesforce Connect and UI integration strategies	10%
Able to describe appropriate trade offs, considerations and alternative options	5%
Able to articulate how correct error handling can be performed for each Salesforce integration pattern	5%

Expected Deliverables

- Please use the following template to guide your presentation. You may add slides, but only if required to further explain your solution. The solution should not have more than 5 slides. The executive who you are presenting to is very busy.
- Please ensure that you only schedule time during the windows s/he has made available

Problem Overview

Bedrock Financial Services has a constant problem with master data across both their customers and their securities/product offerings. Currently, basic customer information is mastered in a customer master which is a hadoop database that is managed by the enterprise architecture group. They have outsourced part of their customer master to an external firm that provides multiple ways to get their data (i.e. real-time, batch, etc). Salesforce will remain the master for all chatter information, opptys, leads and cases. Note that the position master, the system that holds the users positions, account information, and other trading specific info, is located in another system. Given the speed of the market, there is a desire to offer real-time access to info in SFDC.

Volumes

- BFS has about 5 million retail customers
- On average, each customer has two accounts (check / savings and credit card)
- On average, there are 10 transactions per day per account
- To enable informed decision-making, BFS users require access to key insights and indicators from the company's multi-petabyte data warehouse, which contains decades of historical transaction data, directly within the Salesforce platform.
- There are 150 Marketing / Sales Reps and about 500 Customer Service Representatives

Section 1

Marketing / Campaigns

- The marketing team reaches out to prospective customers through email and social media channels targeting certain demographics. They use Marketing Cloud to achieve this. The Marketing team would like to get information on Leads in Near Real-Time so that they can swarm a team immediately to track the lead.
- The details of the prospective customers are captured in a form exposed in a custom portal (Drupal-based) to be followed up by Bedrock sales reps.
- At present the # of prospective customers are in the low hundreds but business is expecting to have the # of prospective customers to increase > 1000+ in two years time.
- The reps are provided a call list of prospective customers based on a highly complex logic. This is based on customer demographics, availability of reps and geographic locations. This logic is a custom business rules engine developed by

BFS. The reps' call lists are maintained in Salesforce and updated daily. The run times are getting longer for updates and needs to be made quicker in order to avoid the risk of missing organization SLA for high-value Customer Prospect Callout.
- The reps would like to have a screen that displays the call list on a map based on the provided address on the exposed form in order to guide them to the most appropriate Bedrock office for an appointment.
- The reps managers will like to get updates in real-time at their Salesforce screens the moment the Rep attends and closes a call with a high priority / high-value customer prospect

Section 2

Customer Conversion / Registration:

- The reps would execute credit checks for the customers through external agencies to qualify them for financial products. The external credit agencies support REST API. The API has to be authenticated and the credit agencies support Openid Connect
- At a given time up to 30 concurrent reps will be using the service for credit checking the customer
- Up to 1 min delay in response expected
- Bedrock Financial will be comparing credit score from at least 3 different agencies. They expect the service to be flexible to expand with additional agencies in future.
- If any of the credit rating agencies are unavailable, then the user must be presented with the error (e.g. Agency XYZ is unavailable). They should also be able to submit the request again. Any re-submitted requests should ensure that only the failed checks are performed.
- Registered customers will be passed to Hadoop DB / SF / Position master near real-time
- From the customer master the community portal user record will be created and activated, an invite will also be sent out to the customer with login and temp password.

Section 3

Customer Management:

- BedRock Financial Services (BFS) have developed a custom web portal based on Drupal. This platform is used for their static web site as well as the customer facing self service portal.
- BFS has leveraged Salesforce Experience Cloud for managing their self service

transaction needs. However, the Communities UI will not be leveraged. The Drupal based portal will need to be able to interact with the Communities solution in real-time.

- The BFS security team have stated that any access to the Communities solution will need to occur via an authenticated end-user (no an API user).
- The transactions that may be performed by customers include (not limited to):
  - Retrieve account address, phone, and other contact information
  - Update address and phone number
  - Retrieve account balances / transaction history. The transaction information is held in their core banking system. Transaction history is available via an Odata connector. Considering the growth rate of business, in a couple of years, especially during holiday seasons, Bedrock financial is expecting that the number of transaction history calls will cross > 100,000 per hour. Bedrock is looking for a strategy that the system is performing seamlessly under such load.
  - Raise an inquiry
- Any updates to customer information needs to be propagated to the Customer Master.
- BFS have also developed a custom (hybrid) mobile app. The functions that are available on the portal are also available to the mobile app.
- The customers may also apply for a loan via the portal. The loan application requires supporting information of various types to be submitted by the customer. The loan application is stored in Salesforce under the 'Application' object and a number of child objects under it. The architecture team is concerned about transaction handling and would like to ensure that no partial records are committed (i.e. if any one of the records in the loan application fails to commit, they would like to roll-back the entire set of records).
- The reps receive calls from the customer for trading specific queries. The reps need to access an external system with complex screen flows and process flows to navigate in order to answer customer queries. Bedrock is looking to leverage this system without swiveling. Bedrock is looking to automate the logging to this system and they are expecting improvement in call handling time and improved customer satisfaction rating with this.

Other requirements

- Mutual Authentication required for all interfaces
- Account for large scale async scenarios with the possibility of millions of events having been generated in a day as part of loan processing.
- The Customer wishes to ensure that all integrations can be centrally monitored. It would be good to have a library of connection templates at hand so that Goto

market speed of integrations becomes faster. Bedrock Financial is looking to create reusable integration assets.

- Where applicable, integration errors should be queued up for an Administrator to correct error and retry
- Prior to go-live, the customer would like to migrate all the data from their existing RockStar system to Salesforce. Post go-live, the customer will keep the RockStar system live for 3 months & few transactions could be done in RockStar too. Therefore, there is a need to keep both systems in sync till the time the RockStar system is fully decommissioned (3 months after go-live)
- Post go-live, the customer would like to sync data of all customers (non-PII fields only) & related transactions from Salesforce to their Enterprise Data Lake which is hosted on a Cloud Platform They would like to sync all changes per record to the data lake such as a) all inserts, b) all updates & c) all deletes. For updates, the history of updates has to be synched with the data lake and not just the last updated values. Also only changed fields have to be synched with data lake and not all the fields